GENAI FINANCIAL FRAUD DETECTION SYSTEM

CAPSTONE PROJECT SUBMISSION

This project presents a Generative AI-powered financial fraud detection system that analyzes banking transactions and identifies suspicious activities using anomaly detection techniques.

The system integrates machine learning models, interactive data visualization, and AI-driven explanations to assist analysts in monitoring potential fraudulent transactions.

Developed using:

Python, Pandas, Scikit-learn, Streamlit, and Generative AI techniques.

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1. PROJECT OVERVIEW

Financial fraud detection is a critical task in modern banking systems. Fraudulent activities such as unauthorized transactions, account takeovers, and suspicious login attempts can cause significant financial losses.

This project develops an intelligent fraud monitoring system using anomaly detection and Generative AI-based reasoning. The system analyzes transaction data and flags unusual patterns that may indicate fraudulent behavior.

The solution consists of four major components:

• Data preprocessing and cleaning

• Anomaly detection using Isolation Forest

• Interactive visualization dashboard

• AI-powered reasoning explaining suspicious transactions

The dashboard allows analysts to explore transaction data, filter suspicious cases, and investigate anomalies in real time.

2. DATA PROCESSING WORKFLOW

The fraud detection system follows a structured workflow to process transaction data and identify anomalies.

Step 1: Data Collection

Transaction data is loaded from a CSV dataset containing transaction ID, transaction amount, and login attempts.

Step 2: Data Cleaning

Missing values are removed and numerical features are converted to appropriate data types.

Step 3: Feature Selection

Important features used for anomaly detection include:

• Transaction Amount

• Login Attempts

Step 4: Data Normalization

The selected features are standardized using StandardScaler to ensure balanced model performance.

Step 5: Anomaly Detection

Isolation Forest is used to identify unusual transaction patterns.

Step 6: Fraud Classification

Transactions predicted as anomalies are labeled as "Suspicious", while others are marked as "Normal".

Step 7: Dashboard Visualization

The processed data is displayed in an interactive Streamlit dashboard for analysis.

3. WORKFLOW DIAGRAM

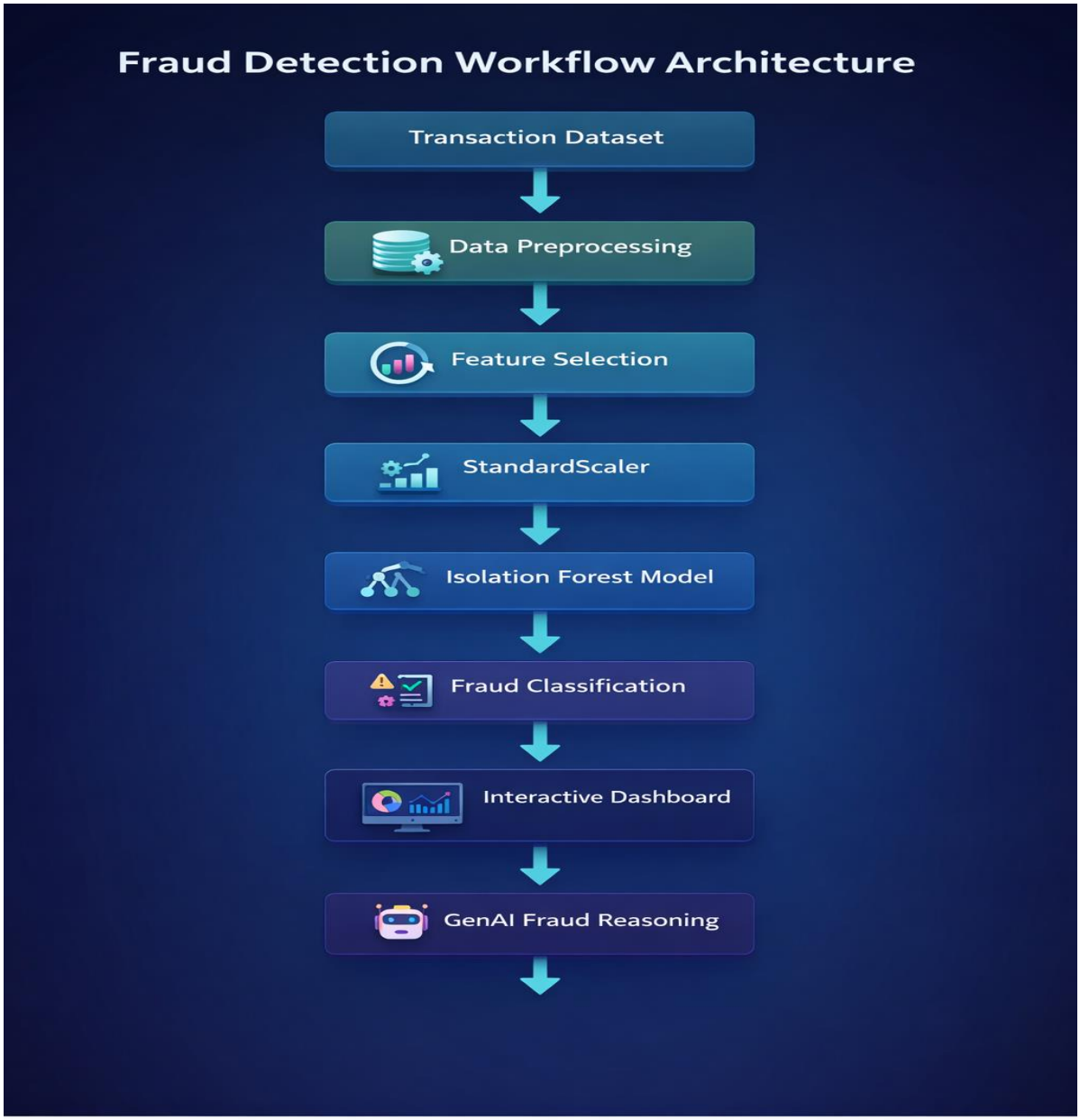

Figure 1: Fraud Detection Workflow Architecture

Transaction Dataset

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Data Preprocessing

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Feature Selection

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StandardScaler

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Isolation Forest Model

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Fraud Classification

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Interactive Dashboard

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GenAI Fraud Reasoning

4. FRAUD DETECTION MODEL LOGIC

The system uses the Isolation Forest algorithm to detect fraudulent transactions.

Isolation Forest is an unsupervised machine learning algorithm designed to identify anomalies in datasets. Unlike traditional classification models, it isolates abnormal data points instead of profiling normal behavior.

The model works by randomly partitioning the data using decision trees. Anomalies require fewer splits to isolate because they differ significantly from the majority of data points.

Key features used in the model include:

• Transaction Amount – unusually large transactions may indicate fraud

• Login Attempts – repeated login attempts may indicate unauthorized access

After training the model, each transaction receives an anomaly score.

Transactions with anomaly score -1 are labeled as:

Suspicious

Transactions with anomaly score 1 are labeled as:

Normal

This approach enables efficient detection of unusual transaction behavior without requiring labeled fraud data.

5. GENAI DASHBOARD

An interactive dashboard was developed using Streamlit to monitor transaction activities.

The dashboard provides the following features:

• Transaction monitoring table displaying all transactions

• Key performance indicators including total transactions, suspicious cases, and normal transactions

• Interactive filtering based on transaction amount

• Visualization of transaction amount distribution

• AI-powered reasoning explaining suspicious transactions

Users can select any transaction from the dashboard and analyze its fraud risk.

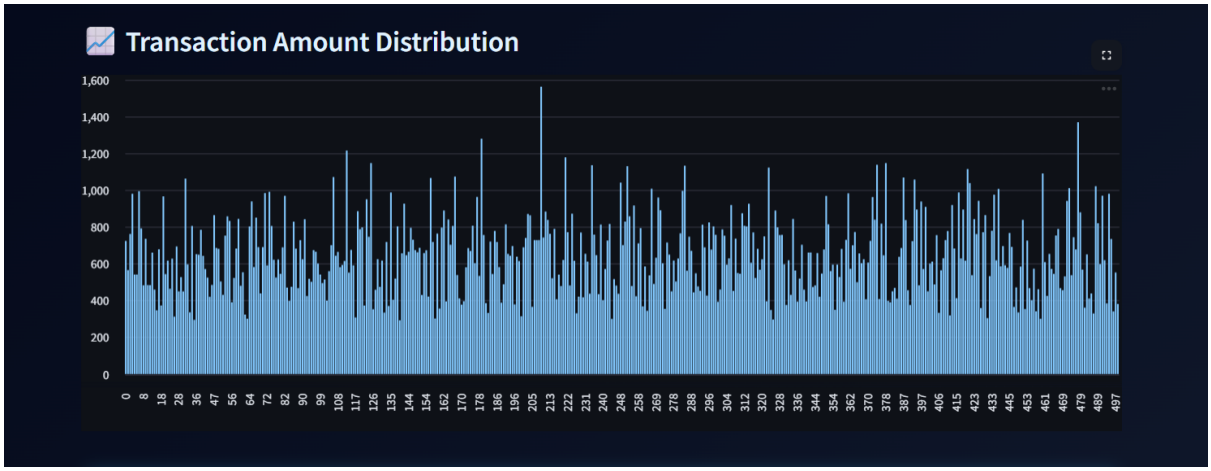
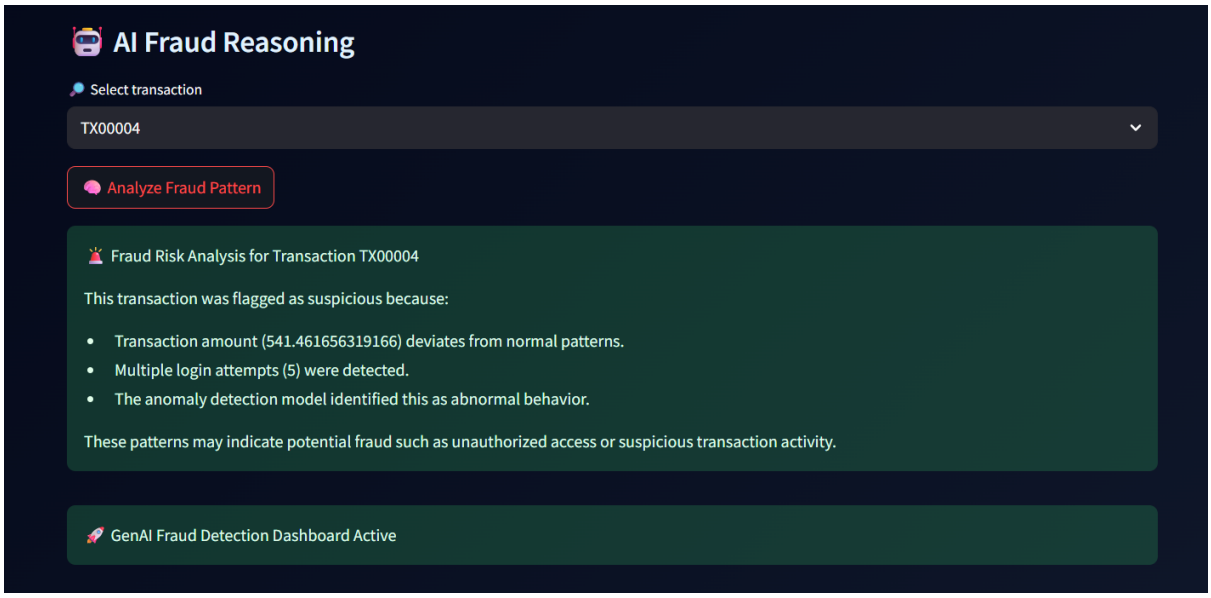
The AI reasoning component provides an explanation describing why the transaction might be considered suspicious based on anomaly detection results.

This improves transparency and helps analysts understand the model's decision.

6. OUTPUT SCREENSHOTS

1.Fraud Detection Dashboard

2.Transaction Amount Distribution Visualization



3.AI Fraud Reasoning Output